Section 2.2 Areas & area ratios

面积与面积比

[AIME-2002-II-Q04] | Part A (Q1-Q5)

Answer: 803

1. The interior contains a central cell and rings of $6, 12, \ldots, 6(n-2)$ cells. Its cell count is $1 + 3(n-2)(n-1)$.

2. At $n = 202$, this is 120601. Each unit hexagon has area $3\sqrt{3}/2$, so $m = 3 \cdot 120601 = 361803$, whose remainder is 803.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2003-I-Q02] | Part A (Q1-Q5)

Answer: 301

1. The green annuli have radii $(1,2),(3,4),\ldots,(99,100)$. Their areas divided by π are $2^2-1^2, 4^2-3^2, \ldots, 100^2-99^2$.

2. Each difference is the sum of the two radii, so the total is $1+2+\cdots+100=5050$. The area ratio is $5050/10000=101/200$, giving 301.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2004-I-Q04] | Part A (Q1-Q5)

Answer: 086

1. For the common corner A, the endpoint triangle is right and its hypotenuse is 2, so its midpoint has distance 1 from A. Conversely the entire corner quarter-circle arises from such endpoints.

2. The central region is the square minus four radius-one quarter-disks, area $4-\pi \approx 0.858407$. Rounding gives $k=0.86$, hence $100k=86$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2004-II-Q01] | Part A (Q1-Q5)

Answer: 592

1. Scale the radius to two. The chord is distance one from the center and subtends 120°. Its smaller circular segment has area $4\pi/3 - \sqrt{3}$.

2. The complementary area is $8\pi/3 + \sqrt{3}$, so the ratio becomes $(8\pi + 3\sqrt{3})/(4\pi - 3\sqrt{3})$.

3. The normalized parameters are $(8, 3, 3, 4, 3, 3)$, whose product is 2592. The remainder is 592.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-II-Q01] | Part A (Q1-Q5)

Answer: 046

1. The four equal angles are $(720-180)/4 = 135^\circ$. For side length s, chords BF, CE separate two right isosceles triangles, each area $s^2/2$, from a rectangle of sides $s, s\sqrt{2}$.

2. The total area is $s^2(1+\sqrt{2})$, so $s^2 = 2116$ and $s = 46$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2008-I-Q02] | Part A (Q1-Q5)

Answer: 025

1. If height $h \leq 10$, triangle area is at most fifty, so $h > 10$. The far square side cuts off a similar triangle with height $h - 10$ and base $10(h - 10)/h$.

2. Overlap area is $5h - 5(h - 10)^2/h = 100 - 500/h$. Set it to eighty to obtain $h = 25$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2014-II-Q03] | Part A (Q1-Q5)

Answer: 720

1. Each eighteen-unit half-bar spans half of the twenty-four-unit separation. Its outward horizontal displacement is $\sqrt{18^2 - 12^2} = 6\sqrt{5}$.

2. The middle width is therefore $a + 12\sqrt{5}$. Two congruent height-twelve trapezoids give total hexagon area $24a + 144\sqrt{5}$.

3. Equate with $36a$ to get $a = 12\sqrt{5}$, hence $a^2 = 720$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2019-I-Q03] | Part A (Q1-Q5)

Answer: 120

1. The triangle is right at R, with area 150. Its other angle sines are $\sin P = 4/5, \sin Q = 3/5$.

2. Remove corner triangles having adjacent sides five and five: their areas are $25(4/5)/2 = 10$, $25(3/5)/2 = 15/2$, and $25/2$. Remaining area $150 - 30 = 120$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2021-II-Q02] | Part A (Q1-Q5)

Answer: 336

1. Let $s = AF$. Triangle AEF is equilateral, so $BD = h = \sqrt{3}(840 - s)/2$. The horizontal distance from E to BD is $h/\sqrt{3}$, giving $[BED] = h^2/(2\sqrt{3})$.

2. $\angle AFG = 120^\circ$, so it must be the apex of the isosceles triangle; hence $FG = FA = s$ and $[AFG] = \sqrt{3}s^2/4$.

3. The ratio gives $3s^2/(2h^2) = 8/9$, or $s = 4h/(3\sqrt{3}) = 2(840 - s)/3$. Thus $s = 336$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2025-I-Q02] | Part A (Q1-Q5)

Answer: 588

1. The side ratios are $AD/AB = AF/AC = 1/7$ and $AE/AB = AG/AC = 5/7$. Therefore $[DEGF] = (25-1)[ABC]/49$, yielding $[ABC] = 588$.

2. Use affine coordinates $A = (0,0), B = (1,0), C = (0,1)$. Then $F = (0,1/7), E = (5/7,0), M = (-1/7,2/7), N = (10/7,-5/7)$.

3. The shoelace cross-product sum for A, F, N, B, C, E, M is $-10/49 + 5/7 + 1 - 5/7 + 10/49 = 1$, exactly that of ABC. Affine area ratios are preserved, so the requested area is 588.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2000-II-Q06] | Part B (Q6-Q10)

Answer: 181

1. If the shorter base is b, the midsegment is $b+50$. The equal heights of the two half-trapezoids give $(2b+50)/(2b+150)=2/3$, so $b=75$ and the other base is 175.

2. Extending the legs to their apex makes similar triangles with area proportional to the square of a parallel section. Equal trapezoid areas imply $x^2-75^2=175^2-x^2$, so $x^2=18125$.

3. The floor of 181.25 is 181.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2001-I-Q09] | Part B (Q6-Q10)

Answer: 061

1. The three corner area fractions are $p(1-r), q(1-p), r(1-q)$. Subtracting them from one yields $[DEF]/[ABC] = 1-(p+q+r)+(pq+qr+rp)$.

2. The pairwise-product sum is $[(2/3)^2 - 2/5]/2 = 1/45$. Thus the area fraction is $1 - 2/3 + 1/45 = 16/45$, giving 61.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2003-II-Q06] | Part B (Q6-Q10)

Answer: 112

1. A triangle’s centroid is one-third of each altitude above the corresponding side. After the half-turn, the opposite vertex lies one-third of the altitude beyond that side.

2. Each of the three exterior corners is therefore similar to the original triangle with scale $1/3$ and area $1/9$ of the original. They are disjoint, so union area is $K + 3K/9 = 4K/3$.

3. Heron gives $K = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84$, hence union area 112.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-I-Q07] | Part B (Q6-Q10)

Answer: 408

1. Let the first-line distance from the vertex be x spacing units. Similar triangles have area proportional to distance squared. Thus A, B, C, D have proportional areas $x^2, 2x+3, 2x+7, 2x+11$.

2. The ratio $(2x+7)/(2x+3) = 11/5$ gives $x = 1/6$. Consequently $[D]/[A] = (2x+11)/x^2 = 408$.

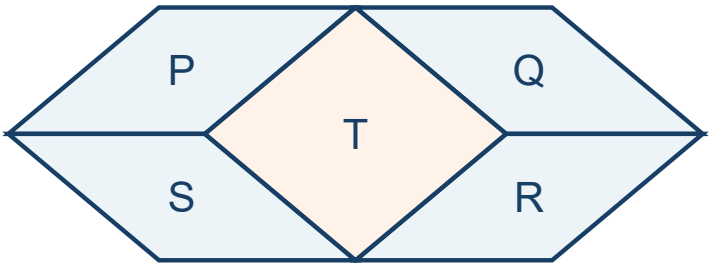
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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-I-Q08] | Part B (Q6-Q10)

Answer: 089



1. Let common side be z and each outer angle incident to the top vertex be α. The central angle is $\pi - 2\alpha$, so $0 < \alpha < \pi/2$ and $z^2 \sin\alpha = \sqrt{2006}$.

2. Central area $K = z^2 \sin 2\alpha = 2\sqrt{2006}\cos\alpha$ takes every value in $(0, \sqrt{8024})$. Since $89^2 < 8024 < 90^2$, the possible integers are 1 through 89.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2011-I-Q08] | Part B (Q6-Q10)

Answer: 318

1. Heron with semiperimeter forty gives area $K = \sqrt{40 \cdot 17 \cdot 13 \cdot 10} = 20\sqrt{221}$. A folded vertex drops by its perpendicular distance to the fold, so a height-h table uses corner triangles of scale factors $ha/(2K), hb/(2K), hc/(2K)$ for opposite sides a, b, c.

2. On any original side, the two endpoint flaps cannot overlap. The three constraints are $h(a+b) \le 2K, h(b+c) \le 2K, h(c+a) \le 2K$. These are also sufficient: the parallel cuts then bound a central tabletop and each folded vertex reaches the floor.

3. The greatest pair sum is $27 + 30 = 57$, so $h_{\max} = 40\sqrt{221}/57$, giving $40 + 221 + 57 = 318$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2013-II-Q10] | Part B (Q6-Q10)

Answer: 146

1. With center O, the points O, B, A are collinear and $AB = 4$. Distances from B, O to any secant through A are in ratio AB/AO, so $[BKL]/[OKL] = 4/(4+\sqrt{13})$.

2. Since $OK = OL = \sqrt{13}$, $[OKL] \leq 13/2$. Equality is possible with a right central angle: the secant need only be at perpendicular distance $\sqrt{13/2}$ from O, which a line through A can attain.

3. Thus maximum $[BKL] = 26/(4+\sqrt{13}) = (104-26\sqrt{13})/3$. The requested sum is 146.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2007-I-Q12] | Part C (Q11-Q15)

Answer: 875

1. The rotation is 15°; denote image vertices $B\prime, C\prime$. Define $D = AB\prime \cap BC$, $E = BC \cap B\prime C\prime$, $F = AC \cap B\prime C\prime$. The overlap is $ADEF = [AB\prime F] - [EB\prime D]$, and $AB\prime \perp BC$.

2. Sine rule in $AB\prime F$ gives $B\prime F = 10\sqrt{6}$, so its area is $50(3+\sqrt{3})$. Right triangle ABD has area fifty. Similarity $EB\prime D \sim ABD$ gives its area as $50[(1-\cos 15^\circ)/\sin 15^\circ]^2 = 50(15 + 8\sqrt{3} - 6\sqrt{6} - 10\sqrt{2})$.

3. Subtract to obtain $500\sqrt{2} - 350\sqrt{3} + 300\sqrt{6} - 600$. Thus the requested expression is $(500+350+300+600)/2 = 875$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2009-I-Q11] | Part C (Q11-Q15)

Answer: 600

1. The fifty points are $P_j = (j, 2009 - 41j)$ for $0 \le j \le 49$. Determinant area is $2009|i - j|/2$.

2. As 2009 is odd, integrality requires equal parity of i, j. There are twenty-five indices of each parity; choose unordered pairs: $2\binom{25}{2} = 600$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2010-I-Q13] | Part C (Q11-Q15)

Answer: 069

1. The radius is 126. With O the midpoint of AB, $AN = AO = ON$, so AON is equilateral. If P is the foot from N to AB, then $AP = 63, NP = 63\sqrt{3}, UP = 21, UO = 42$.

2. Let Q be the foot from U to CD. The region on the A side of the cutting line has one-third of the semicircle minus $[NUO]$, plus one-third of the rectangle plus $[UQT]$. Its rectangle portion is at most two-thirds (its widths are 84 and $DT \leq 252$), and its semicircle portion is less than one-third; hence this side cannot be the larger two-thirds portion.

3. It must be exactly one-third, so $[NUO] = [UQT]$. Relative to $[NUP]$, the first area has ratio $UO/UP = 2$, while the similar triangle UQT has area ratio $(UQ/NP)^2$.

4. Thus $DA = UQ = \sqrt{2}\,NP = 63\sqrt{6}$, giving 69. The resulting T lies on CD, since $DT = 84 + 21\sqrt{2} < 252$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2025-II-Q14] | Part C (Q11-Q15)

Answer: 104

1. First verify the angular order. Since $AK = AL = KL$, the rays AK, AL differ by 60°. If AL lay below AK inside the right angle, both $\angle BAK$ and $\angle LAC$ would exceed 60°, so the equal-radius isosceles relations would give $AB, AC < 14$, contrary to $BC = 38$. Hence the order is AB, AK, AL, AC.

2. AKL is equilateral. Put $\alpha = \angle BAK, \beta = \angle LAC$, so $\alpha + \beta = 30^\circ$. From the two isosceles triangles, $AB = 28\cos\alpha, AC = 28\cos\beta$.

3. Pythagoras gives $\cos^2\alpha + \cos^2\beta = 361/196$, whence $\cos(\alpha - \beta) = 55\sqrt{3}/98$. Therefore $[ABC] = 392\cos\alpha\cos\beta = 208\sqrt{3}$.

4. The removed areas satisfy $[ABK] + [ACL] = 98(\sin 2\alpha + \sin 2\beta) = 55\sqrt{3}$ and $[AKL] = 49\sqrt{3}$. Subtract to obtain $[BKLC] = 104\sqrt{3}$, so $n = 104$.

Source / 来源